

Multiple Choice (10 marks)

1. What is the significance of the 1:2:4 resonance in the Jupiter's moons system?
 - (a) The resonance pulls Io in different directions and generates heat.
 - (b) It makes the orbit of Io slightly elliptical.
 - (c) It creates a gap with no asteroids between the orbits.
 - (d) It prevents formation of the ring material into other moons.
2. In addition to the conditions required for any solar eclipse what must also be true in order for you to observe a total solar eclipse?
 - (a) The Earth must lie completely within the Moon's penumbra.
 - (b) The Moon's penumbra must touch the area where you are located.
 - (c) The Earth must be near aphelion in its orbit of the Sun.
 - (d) The Moon's umbra must touch the area where you are located.
3. Approximately how long does it take Pluto to orbit the Sun once?
 - (a) 150 years
 - (b) 200 years
 - (c) 250 years
 - (d) 300 years
4. Which factor is most important in determining the history of volcanism and tectonism on a planet?
 - (a) size of the planet
 - (b) presence of an atmosphere
 - (c) distance from the sun
 - (d) rotation period
5. Which of the following is/are NOT caused by orbital resonance?
 - (a) 2:3 periodic ratio of Neptune:Pluto
 - (b) Kirkwood Gaps.
 - (c) Gaps in Saturn's rings.
 - (d) Breaking of small Jovian moons to form ring materials.

True / False (10 marks)

Answer True or False

6. True or False: The Cassini division is the most opaque ring of Saturn, composed of dense, reflective ice particles.
7. True or False: The presence of life on Earth is not a property that formation theories of the solar system need to explain.
8. True or False: The Moon does not rotate, which is why we see the same face at all times.
9. True or False: Saturn and Jupiter do not both exhibit equatorial wind speeds exceeding 900 miles per hour in their atmospheres.
10. True or False: The Arecibo Telescope had the largest parabolic antenna among radio telescopes before its decommissioning in 2020.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss why the term "gas giants" is considered misleading when describing Jupiter and the other jovian planets, focusing on their composition and the implications for planetary science.
12. Discuss the significance of hydrogen as the most abundant element in the composition of Jovian planets and analyze its implications for their formation and atmospheric characteristics.

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